

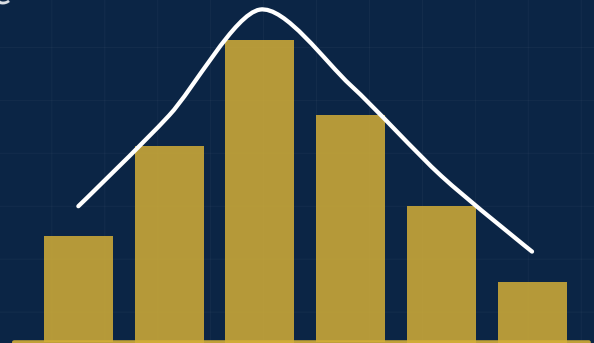


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CHAPTER 6 • STATISTICS & PROBABILITY • TOPIC 6

Combined & Conditional Probability

AND multiplies, OR adds. “At least one” is 1 minus “none”. “Given that” shrinks the whole to the group you are told about.



4

VARIANTS

7

PAST-PAPER PARTS

44

PRACTICE QUESTIONS

YOU WILL MASTER

V1 AND / OR

V2 Without replacement & “at least one”

V3 “Given that”

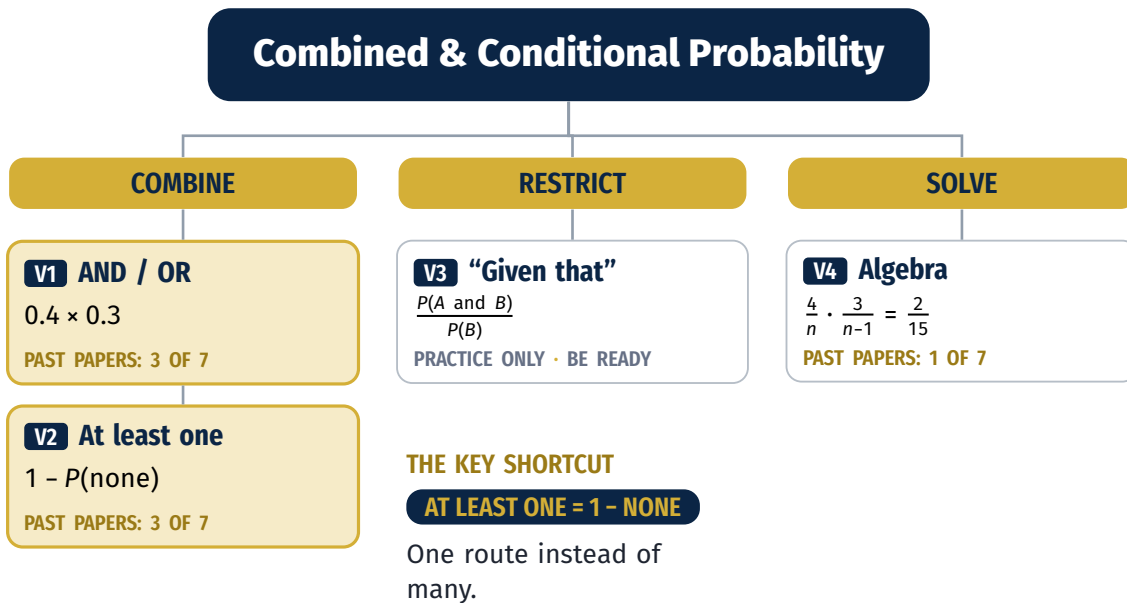
V4 Algebra in probability

TOPIC MAP · EVERY VARIANT ON ONE PAGE

Combined probability at a glance



Your notes



★ How often does each variant come up?

7 past-paper questions (Jan 2020 → Nov 2025); 44 practice parts — grouped by type. Trees and Venn diagrams are in Topic 5.

		PAST PAPERS (OUT OF 7)		PRACTICE PARTS	
V1	AND and OR	●●●●●●●	3	●●●●●●●●●●	13
V2	Without replacement, at least one	●●●●●●●	3	●●●●●●●●●●●●●●●●	16
V3	Given that	●●●●●●●	0	●●●●●	5
V4	Algebra in probability	●●●●●●●	1	●●●●●●●●●●	10

WHAT THIS TELLS YOU One late question (Q16–22, 3–4 marks) with no diagram given: three picks, or “at least one”. Use the complement — it turns a long list of routes into one line.

BEFORE YOU START

The big idea

**AND multiplies, OR adds.**

Both things happen (one after the other): **multiply**. Either of two outcomes that can't both happen: **add**.

Lots of routes? Work out the **opposite** and take it from 1.

at least one even

1 - none even

$$1 - \left(\frac{1}{2}\right)^4 = \frac{15}{16}$$

 The rules
AND

$$P(A) \times P(B)$$

Independent events.

OR

$$P(A) + P(B)$$

Can't both happen.

Not

$$1 - P(A)$$

Complement.

Given B

$$\frac{P(A \text{ and } B)}{P(B)}$$

The group you're told about.

 Without replacement
First pick

$$\frac{3}{12}$$

From the full bag.

Second pick

$$\frac{2}{11}$$

One fewer of that kind and in total.

Third pick

$$\frac{1}{10}$$

And again.

**CALCULATOR FIRST, THEN CHECK**

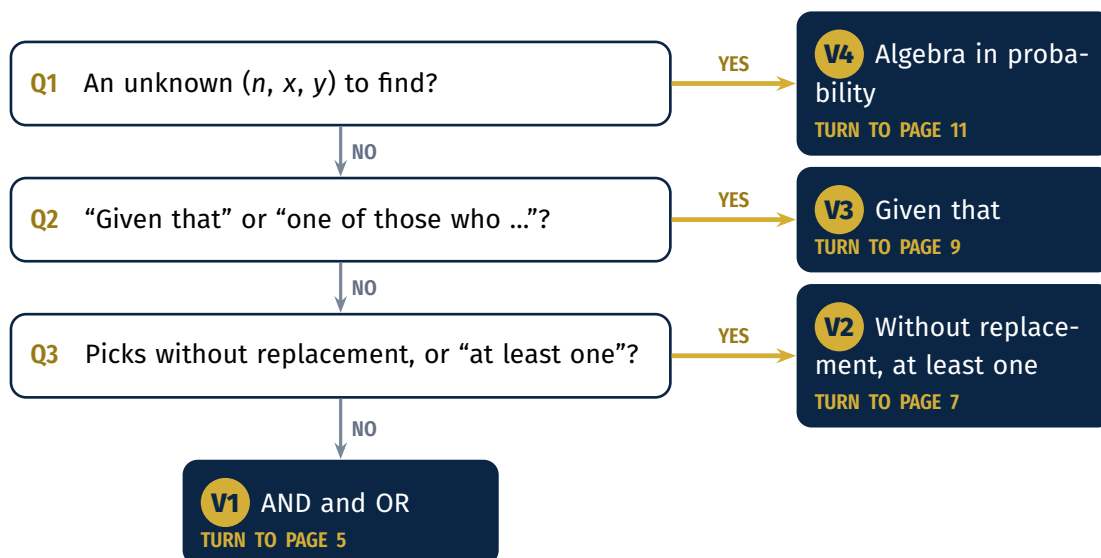
Exact fractions with the fraction key; simplify at the end.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD



What is it?



When you see → you do



How do I do it?



Watch out



Worked example



Exam tip



Check yourself



Calculator check

V1**VARIANT 1** MOST COMMON IN THE EXAM**AND & OR**

Independent: multiply for AND; add the separate routes for OR; use $P(\text{both})$ to undo.

SEEN IN PAST PAPERS

●●●●●●● **3/7**

PRACTICE QUESTIONS: 13



Your notes

Came up in: Jan 2020 P1HR Q16 • Jun 2022 P2HR Q14 • Nov 2021 P2H Q16

? What is it?**In plain words**

Independent events:

 $P(A \text{ and } B) = P(A) \times P(B)$. Given $P(\text{both})$ and one probability? **Divide** to get the other.

“Same colour”: add the routes RR + GG + YY.

$$0.6 \times P(C) = 0.42$$

$$P(C) = 0.7$$

$$\text{neither: } 0.4 \times 0.3$$

IT LOOKS LIKE

win both 0.42

same colour

How do I do it?**1 Write**

Each event's probability (and its “not”).

2 AND

Multiply along.

3 OR

Add the routes.

4 UndoFrom $P(\text{both})$, divide.**AND $\times$ OR +** Independent events.

V1 AND & OR

VARIANT 1 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“the probability that she will win both games is 0.42” → $0.6 \times P = 0.42$

“will not win either game” → $P(\text{not}_1) \times P(\text{not}_2)$

“both have the same colour” → **Sum of each colour squared (with replacement)**

WORKED EXAMPLE **PAST PAPER**

January 2020 · Paper 1HR · Q16 · 4 marks

$P(\text{Steffi wins tennis}) = 0.6$; $P(\text{wins both tennis and chess}) = 0.42$. Find $P(\text{she wins neither game})$.

STEP 1 $0.6 \times P(\text{chess}) = 0.42$ ◀ AND

STEP 2 $P(\text{chess}) = 0.7$ ◀ divide

STEP 3 $P(\text{not}_T) = 0.4$, $P(\text{not}_C) = 0.3$ ◀ complements

STEP 4 $0.4 \times 0.3 = 0.12$ ◀ neither

ANSWER 0.12✓ $0.42 + 0.18 + 0.28 + 0.12 = 1$ ✓**WATCH OUT**

$$\times 1 - 0.42 = 0.58$$

$$\checkmark 0.4 \times 0.3$$

“Not both” is not the same as “neither”.

EXAM TIP

List all four outcomes and check they add to 1.

Check yourself

calculator first · answers on page 13

1.1 Jun 2022 P2HR

Four fair dice are thrown. $P(\text{at least one even})$?

Answer:

1.2 Nov 2021 P2H

15 counters: 4 red, 5 green, 6 yellow. Two picks with replacement. $P(\text{same colour})$?

Answer:

1.3 Dr Eslam

$P(A) = 0.3$, $P(B) = 0.5$, independent. Find $P(A \text{ or } B)$.

Answer:

V2**VARIANT 2** MOST COMMON IN THE EXAM

Without Replacement & “At Least One”

Shrink the bag each pick; “at least one”
= 1 – “none”.

SEEN IN PAST PAPERS

●●●●●●●● **3 / 7**

PRACTICE QUESTIONS: 16


Your notes

Came up in: Jun 2023 P2H Q20 • May 2024 P1H Q16 • Nov 2025 P1H Q22

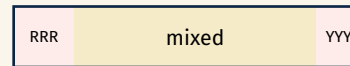
? What is it?

In plain words

Three picks without replacement:

$$\frac{15}{20} \times \frac{14}{19} \times \frac{13}{18}$$

“At least one of each colour” = 1 – (all red) – (all yellow). “At least one red still in the bag” = 1 – (all reds taken).



$$\text{mixed} = 1 - \text{RRR} - \text{YYY}$$

IT LOOKS LIKE

3 sweets

at least one of each

How do I do it?

1 Opposite

Is “none” / “all the same” quicker?

2 Picks

Fractions shrinking each time.

3 Multiply

Along each route.

4 Complement

1 – the routes you don’t want.

AT LEAST ONE = 1 – NONE One line instead of many.

V2 Without Replacement & “At Least One”

VARIANT 2 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“at least one sweet of each colour” → 1 - all red - all yellow

“at least one red counter is still in the bag” → 1 - all reds taken

“takes at random 3 ...” → No replacement: 20, 19, 18

WORKED EXAMPLE **PAST PAPER**

May 2024 · Paper 1H · Q16 · 4 marks

20 sweets: 15 red, 5 yellow. Fred takes 3. Find the probability he takes at least one sweet of each colour.

STEP 1 $P(RRR) = \frac{15}{20} \cdot \frac{14}{19} \cdot \frac{13}{18} = \frac{2730}{6840}$ ◀ all red

STEP 2 $P(YYY) = \frac{5}{20} \cdot \frac{4}{19} \cdot \frac{3}{18} = \frac{60}{6840}$ ◀ all yellow

STEP 3 $1 - \frac{2790}{6840} = \frac{4050}{6840}$ ◀ complement

STEP 4 $= \frac{45}{76}$ ◀ simplify

ANSWER $\frac{45}{76}$

✓ between 0 and 1 ✓

WATCH OUT

× listing RRY, RYR, ... (6 routes)

✓ 1 - RRR - YYY

Same answer, far less work.

EXAM TIP

Keep one common denominator
(20 · 19 · 18) for all routes.

Check yourself

calculator first · answers on page 13

2.1 Jun 2023 P2H

12 counters: 3 red, 9 green. Three are taken. P(at least one red still in the bag)?

Answer:

2.2 Nov 2025 P1H

15 buttons: 3 red, 2 pink, 10 blue. Three are taken. P(at least one pink still in the box)?

Answer:

2.3 Dr Eslam

3 red, 5 blue counters. Three taken without replacement. P(all blue)?

Answer:

V3

VARIANT 3

**“Given That”:
Conditional
Probability**

The group you are told about becomes the whole: $P(A | B) = \frac{P(A \text{ and } B)}{P(B)}$.

SEEN IN PAST PAPERS

● ● ● ● ● ● ● 0 / 7

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 5


 Your notes
? What is it?**In plain words**

“Given that he was late” — look **only** at the late routes. The answer is (the route you want) ÷ (all the late routes).

With counts: $\frac{\text{in both groups}}{\text{in the given group}}$.

IT LOOKS LIKE

given late

one of the footballers

rain, late: $0.25 \times 0.3 = 0.075$ dry, late: $0.75 \times 0.65 = 0.4875$

$$P(\text{rain} | \text{late}) = \frac{0.075}{0.5625}$$

How do I do it?**1 Given group**

All routes (or people) in it.

2 Its probability

Add those routes.

3 Wanted part

The route that is also what you want.

4 Divide

Wanted ÷ given.

GIVEN = NEW WHOLE Divide by it.

V3 “Given That”: Conditional Probability

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Given that he was late, find the probability it rained”

→ **Rain-and-late ÷ all late**

“A footballer is chosen. $P(\text{swims})$ ”

→ **Footballers who swim ÷ footballers**

“If ... then ...” probabilities

→ **Already conditional: use them along branches**

**WORKED EXAMPLE** **PRACTICE**

Practice • Medium • Q14 (extended) • 4 marks

$P(\text{rain}) = 0.25$. If it rains, $P(\text{late}) = 0.3$; if not, $P(\text{late}) = 0.65$. (a) Find $P(\text{late})$. (b) Given that he is late, find $P(\text{it rained})$.

STEP 1 $0.25 \times 0.3 = 0.075$

◀ rain and late

STEP 2 $0.75 \times 0.65 = 0.4875$

◀ dry and late

STEP 3 $P(\text{late}) = 0.5625$

◀ (a) add

STEP 4 $\frac{0.075}{0.5625} = \frac{2}{15}$

◀ (b) divide

ANSWER (a) 0.5625 (b) $\frac{2}{15}$ ✓ smaller than 0.25: rain makes him less late ✓

**WATCH OUT**

✗ answer 0.075

✓ $0.075 \div 0.5625$

“Given late” — divide by $P(\text{late})$.

**EXAM TIP**

Draw the tree first; then circle the routes in the “given” group.

**Check yourself**

calculator first • answers on page 13

3.1

Dr Eslam

40 students: 25 play football; 10 of those also swim. A footballer is chosen. $P(\text{swims})$?

Answer:

3.2

Dr Eslam

35 boys (14 cycle) and 25 girls (6 cycle). A cyclist is chosen. $P(\text{girl})$?

Answer:

3.3

Practice • Medium

$P(\text{pass first try}) = 0.7$; if fail, $P(\text{pass the second try}) = 0.9$. $P(\text{passes})$?

Answer:

V4

VARIANT 4

Algebra in Probability

Write the probability with the letter, set it equal to the value given, solve (often a quadratic).

SEEN IN PAST PAPERS

● ● ● ● ● ● ● 1/7

PRACTICE QUESTIONS: 10



Your notes

Came up in: May 2023 P1HR Q12

? What is it?

In plain words

Unknown number of counters n :
write each pick as a fraction with n ,
multiply, set equal to the given
probability, **cross-multiply**.

Usually a **quadratic** — keep the
positive whole answer.

$$\frac{4}{n} \cdot \frac{3}{n-1} = \frac{2}{15}$$

$$180 = 2n(n-1)$$

$$n^2 - n - 90 = 0$$

$$n = 10$$

IT LOOKS LIKE

$$\frac{4}{n} \cdot \frac{3}{n-1} = \frac{2}{15}$$

$$x + y = 0.19$$

How do I do it?

1 Fractions

Each pick with the letter.

2 Equation

Product (or sum) = given.

3 Clear

Cross-multiply; tidy to = 0.

4 Solve

Positive whole answer; check it.

LETTER IN, EQUATION OUT

Reject negatives and fractions.

V4 Algebra in Probability

VARIANT 4 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“A bag contains n beads ... find n ” → **Probability in n = given**

“Show that $x + y = 0.19$ ” → **All probabilities add to 1**

“Show clear algebraic working” → **The equation and its factorised form**

**WORKED EXAMPLE DR ESAM**

Algebra · 4 marks

A bag has n counters, 4 of them red. Two are taken without replacement. $P(\text{both red}) = \frac{2}{15}$. Find n .

STEP 1 $\frac{4}{n} \times \frac{3}{n-1} = \frac{2}{15}$ ◀ two picks

STEP 2 $180 = 2n(n-1)$ ◀ cross-multiply

STEP 3 $n^2 - n - 90 = 0 \Rightarrow (n-10)(n+9) = 0$ ◀ quadratic

STEP 4 $n = 10$ ◀ positive

ANSWER $n = 10$

$$\checkmark \frac{4}{10} \cdot \frac{3}{9} = \frac{2}{15} \checkmark$$

**WATCH OUT**

$$\times \frac{4}{n} \cdot \frac{3}{n}$$

$$\checkmark \frac{4}{n} \cdot \frac{3}{n-1}$$

One fewer in the bag on the second pick.

**EXAM TIP**

Substitute your n back in — it should give exactly the probability in the question.

**Check yourself**

calculator first · answers on page 13

4.1

May 2023 P1HR

Dice: 1: x , 2: 0.15, 3: 0.5,
4: y , 5: 0.13, 6: 0.03.
 $3x - y = 0.09$. Find x
and y .

Answer:

4.2

May 2021 P2H

N pens, black = red + 3.
 $P(\text{black then red}) = \frac{9}{35}$.
Find N .

Answer:

4.3

Dr Eslam

x red and 5 blue. $P(\text{two red, no replacement}) = \frac{1}{6}$. Find x .

Answer:

CHECK YOURSELF

Answers



Every answer below was checked with sympy (check_answers.py).

1.1 $\frac{15}{16}$

1.2 $\frac{77}{225}$

1.3 0.65

2.1 $\frac{219}{220}$

2.2 $\frac{34}{35}$

2.3 $\frac{5}{28}$

3.1 $\frac{2}{5}$

3.2 $\frac{3}{10}$

3.3 0.97

4.1 $x = 0.07, y = 0.12$

4.2 $N = 15 \text{ or } 21$

4.3 4 red

THE WHOLE TOPIC ON ONE PAGE

Combined probability cheat sheet



Your notes
V1 AND / OR 3/7

0.4×0.3

AND $\times$, OR $+$.**V2 At least one 3/7**

$1 - RRR - YYY$

Complement; shrink the bag.

V3 Given that

$$\frac{0.075}{0.5625}$$

Given group = new whole.

V4 Algebra 1/7

$n^2 - n - 90 = 0$

Equation; positive whole n .**TOP 3 LOST MARKS**

1. "not both" for "neither" $\times$
2. same bottom on every pick $\times$
3. not dividing for "given" $\times$

BOTTOM LINE AND $\times$ · OR $+$ · at least one = $1 - \text{none}$ · given = new whole.